Data Mining: Introduction to Data Mining

Spring 2026 (7761)

Dept. of Computer Science and Engineering
Instructor: Ting-Wei Lin

2026/09/11, Week 1

Course Introduction

Data Mining

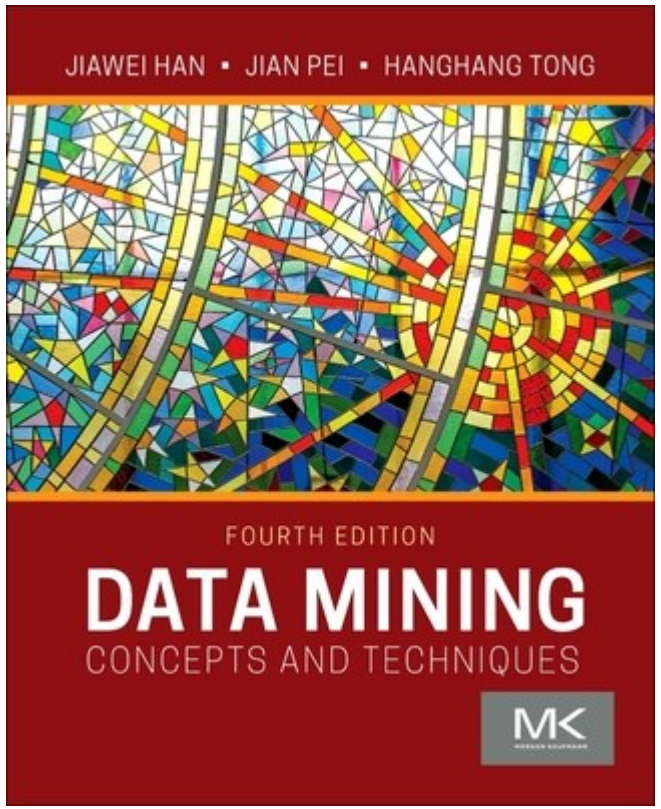
Course Information

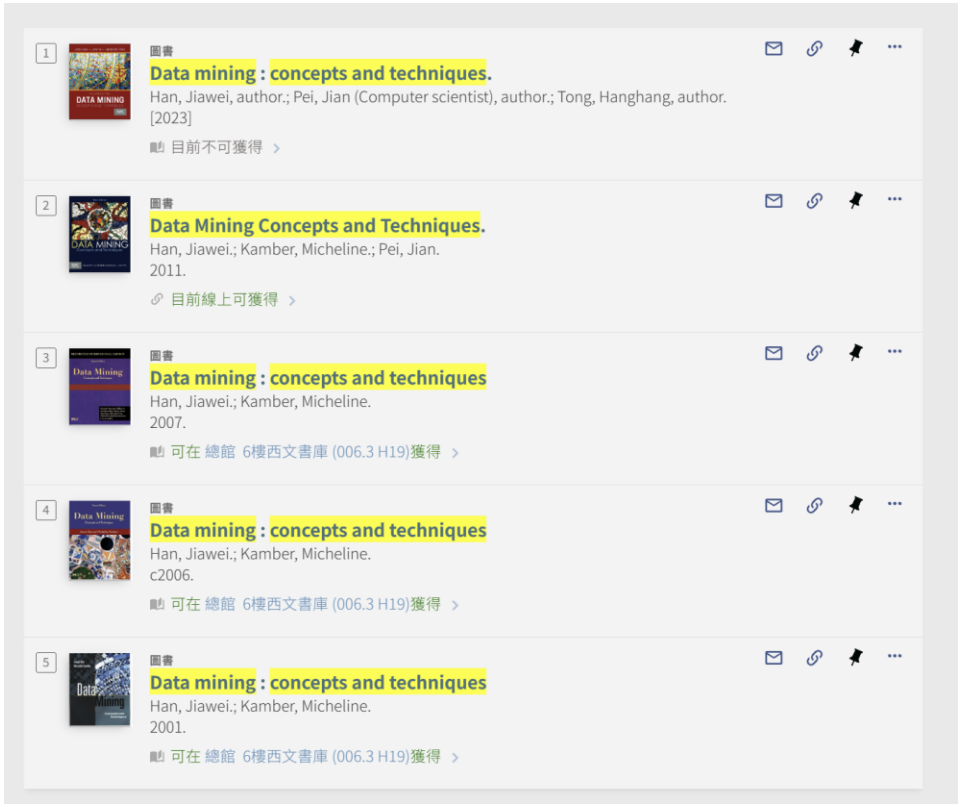
- Instructor: 林鼎崴 Ting-Wei Lin
- Email: tingwei.lin@cs.nchu.edu.tw
- Teaching Assistant:
  - 陳鉦元
  - 吳卓桐
- Lecture Time: Friday 18:20 – 21:00
- Office hour: appointment by email
- Grading
  - Exam: 25%
  - Assignment: 60% (20% for each)
  - Paper Reading: 15% (video recording)

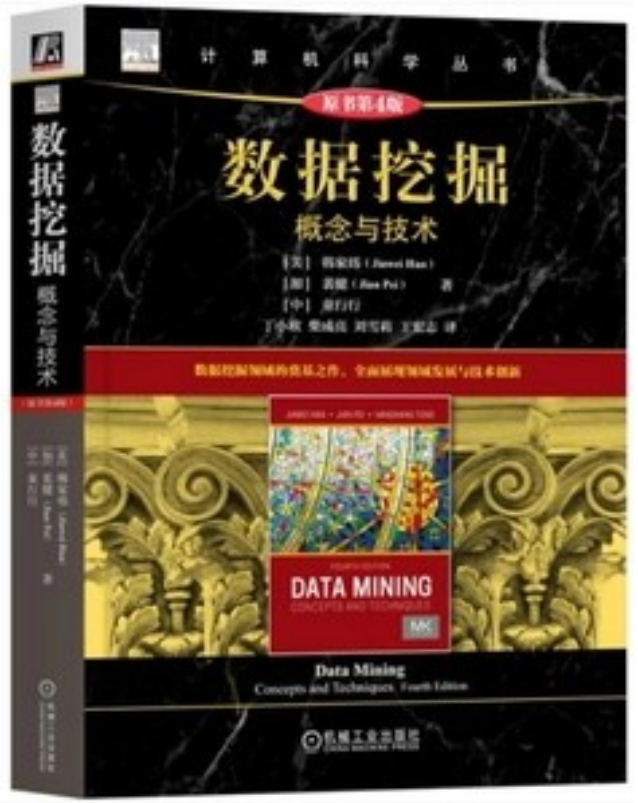
Textbook

- Data Mining: Concepts and Techniques, 4/e

  Jiawei Han , Jian Pei , Hanghang Tong



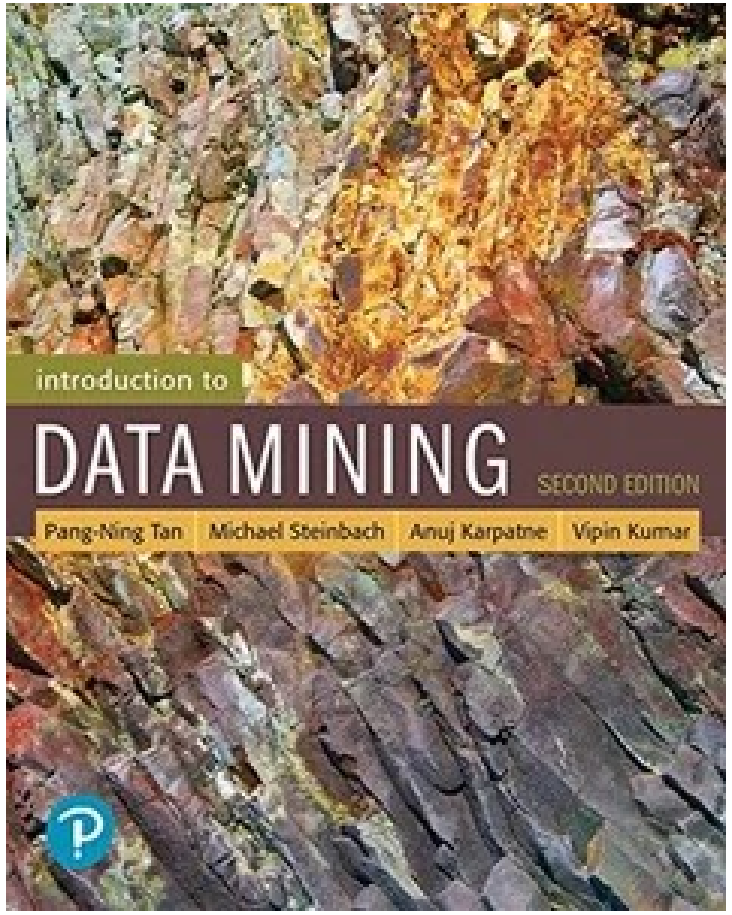


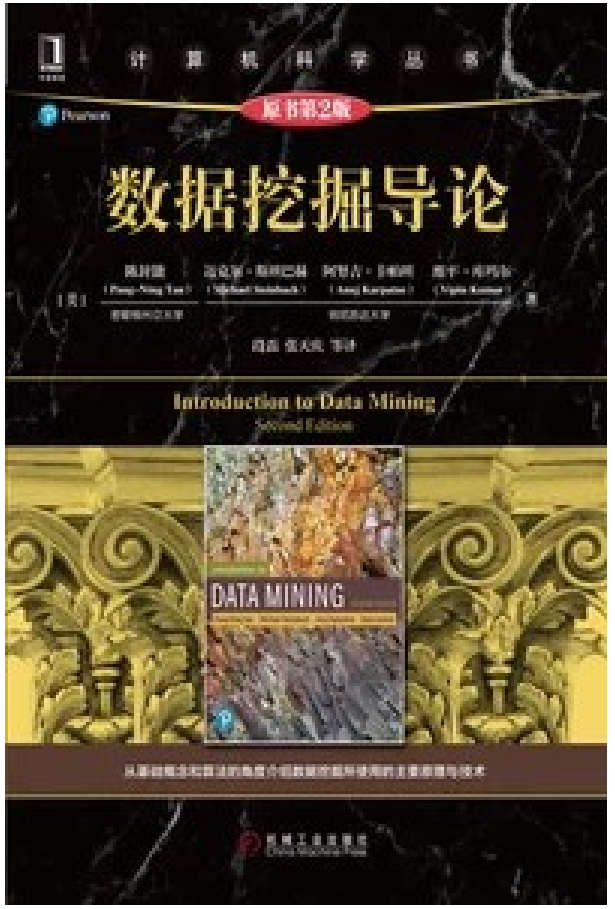


Reference book

- Introduction to Data Mining 2/e

  Pang-Ning Tan, Michael Steinbach, Anuj Karpatne, Vipin Kumar





Syllabus

Week 1 (09/11):

Introduction to Data Mining

Week 2 (09/18):

Data Preprocessing

Week 3 & 5 (09/25 & 10/09):

No Class (中秋節、國慶日)

Week 4 & 6 (10/02 & 10/16):

Association Rule Mining

Assignment 1: Weka (Deadline: 11/06)

Syllabus (cont.)

Week 7 – 9 (10/23 – 11/06):

Classification

Assignment 2: LibSVM (Deadline: 11/27)

Week 10 - 11 (11/13 – 11/20):

Clustering

Assignment 3: Kaggle Practice (Deadline: 12/18)

Week 12 (11/27):

Exam Week

Syllabus (cont.)

Week 13 (12/04):

Python Tools for Data Mining

Week 14 (12/11):

Outlier Detection

Week 15 (12/18):

Advanced Topic:

a) Recommendation
b) Social Network Mining

Week 16 (12/25):

Paper Reading (video recording)

Important Dates

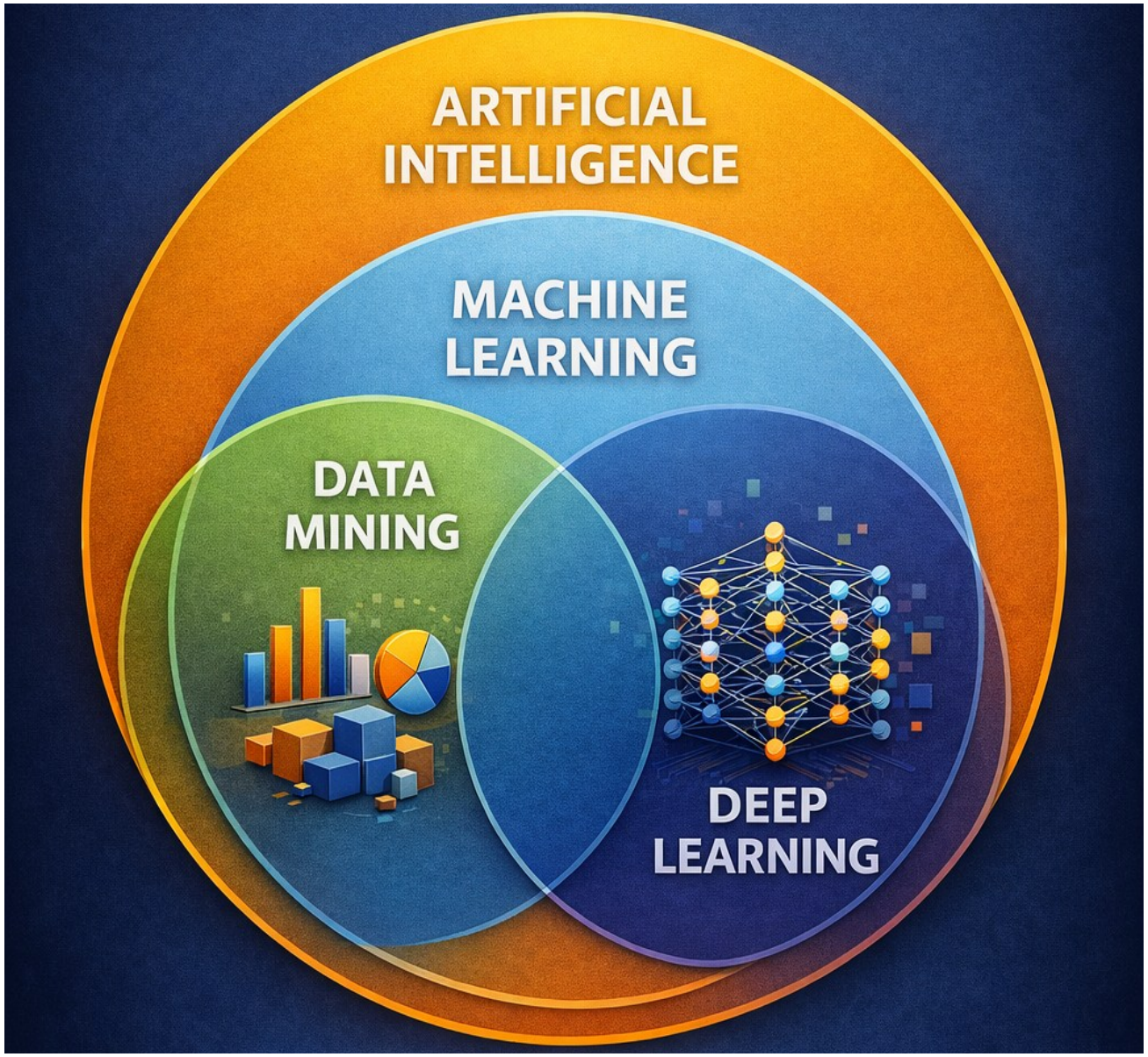
- Assignment - 60%
  - Assignment 1: Weka (10/16 - 11/06) [20%]
  - Assignment 2: LibSVM (11/06 - 11/27) [20%]
  - Assignment 3: Kaggle Practice (11/20 - 12/18) [20%]
- Exam - 25% (11/27)
- Paper Reading - 15% (12/25)

Introduction to Data Mining

Data Mining Journal and Conferences

	Publikation	h5-indeks	h5-median
1.	IEEE Transactions on Knowledge and Data Engineering	126	209
2.	ACM SIGKDD Conference on Knowledge Discovery and Data Mining	124	212
3.	International Conference on Artificial Intelligence and Statistics	101	156
4.	ACM International Conference on Web Search and Data Mining	86	135
5.	Journal of Big Data	85	176
6.	ACM Conference on Recommender Systems	57	90
7.	Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery	56	134
8.	IEEE International Conference on Big Data	55	89
9.	ACM Transactions on Intelligent Systems and Technology	52	110
10.	Data Mining and Knowledge Discovery	51	81
11.	Knowledge and Information Systems	51	73
12.	IEEE International Conference on Data Mining	49	86
13.	ACM Transactions on Knowledge Discovery from Data	49	76
14.	Social Network Analysis and Mining	47	72
15.	Big Data Mining and Analytics	45	88
16.	European Conference on Machine Learning and Knowledge Discovery in Databases	41	84
17.	International Journal of Data Science and Analytics	39	64
18.	SIAM International Conference on Data Mining (SDM)	36	66
19.	Online Social Networks and Media	33	54
20.	Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD)	33	49

① Artificial Intelligence
② Machine Learning
③ Data Mining
④ Deep Learning



① Artificial Intelligence

- A technological domain that empowers computers and machines to emulate human learning, understanding, problem-solving, decision-making, creativity, and autonomy.

② Machine Learning

- Learning from data to make predictions/decisions.

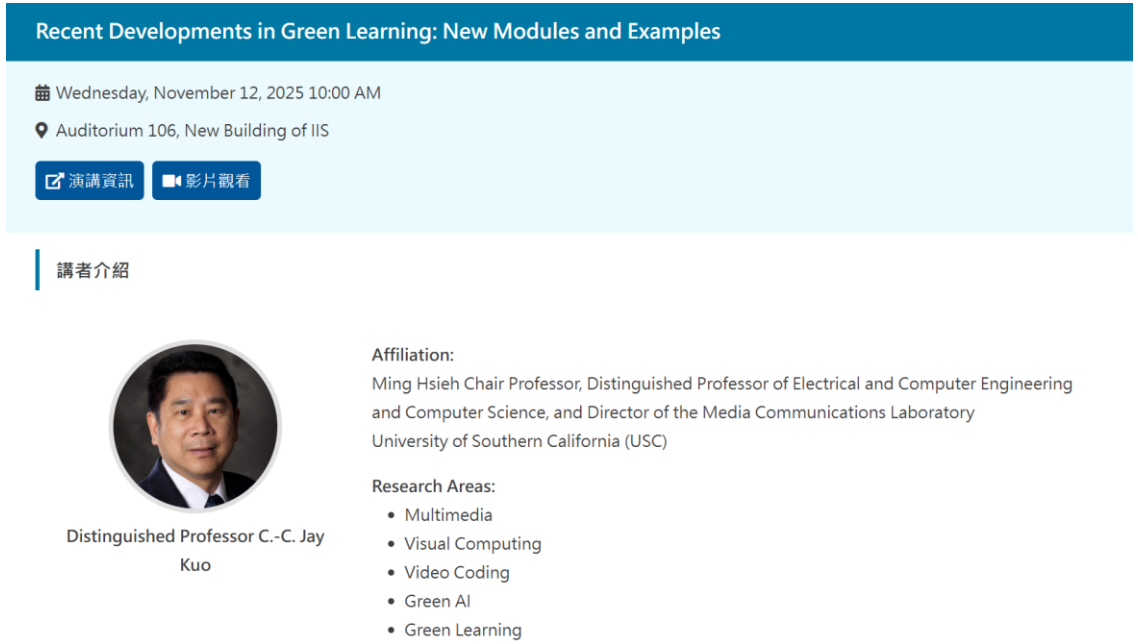
③ Deep Learning

- Machine learning using deep neural networks.

④ Data Mining?

Green Learning

- A sustainable AI paradigm focusing on low-carbon, efficient models.
- 中研院院士: 郭宗杰
  - https://academicians.sinica.edu.tw/index.php?r=academician-n%2Fshow&id=776



https://opendls.iis.sinica.edu.tw:8443/vs/sharing/jEiphXYx#!aG9tZV92aWRlby0yMDY=

What is Data Mining

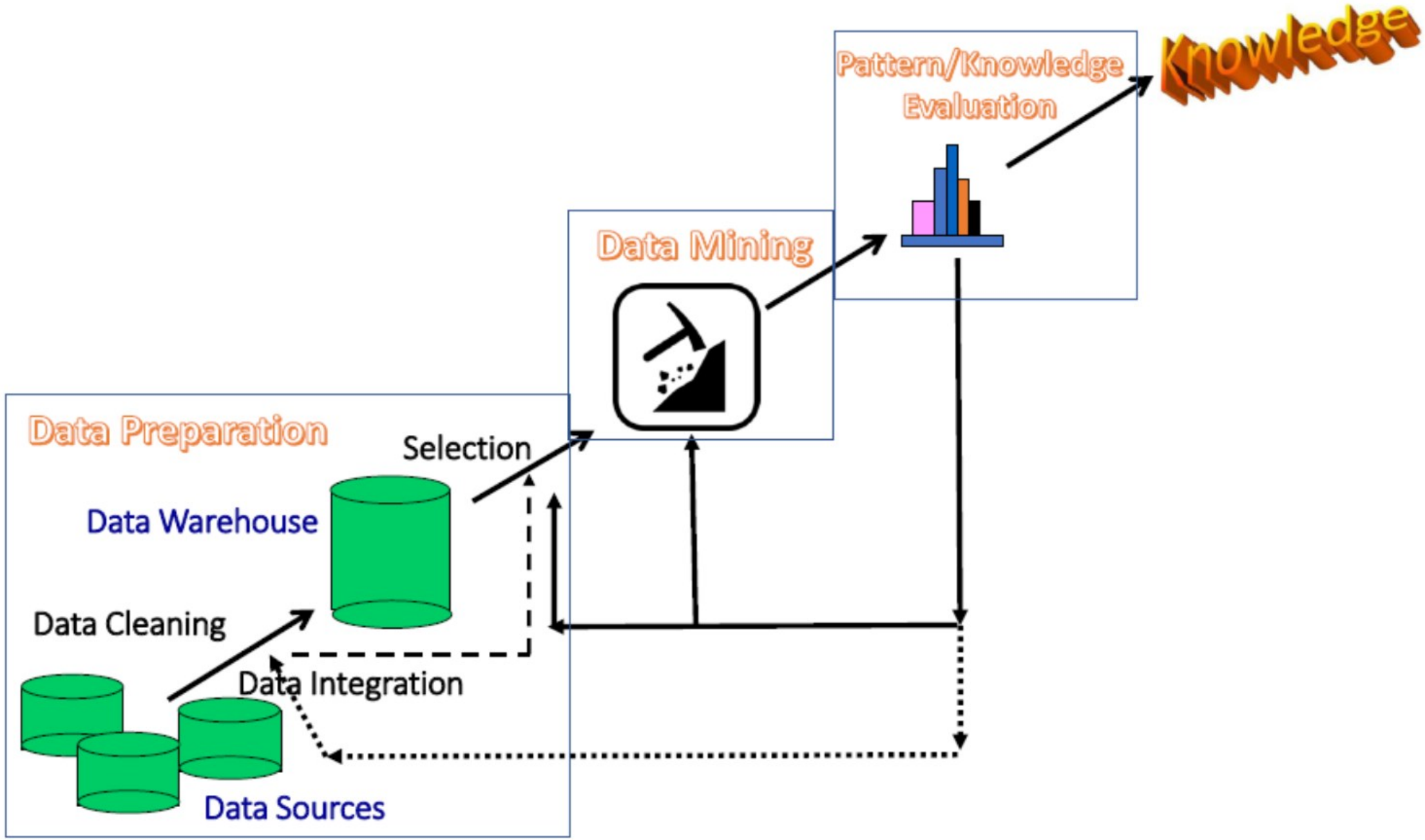
- Discover interesting patterns, models, and other kinds of knowledge in large data sets.
  - Gold nuggets from data, appeared in 1990s
  - Knowledge mining from data, KDD (i.e., Knowledge Discovery from Data)
  - Pattern discovery
  - Knowledge extraction
  - Data archaeology
  - Data analytics
  - Information harvesting

Knowledge Discovery Process

1. Data preparation
   - Data cleaning: to remove noise and inconsistent data.
   - Data integration: where multiple data sources may be combined.
   - Data transformation: where data are transformed and consolidated into forms appropriate for mining by performing summary or aggregation operations.
   - Data selection: where data relevant to the analysis task are retrieved from the database.

Knowledge Discovery Process (cont.)

2. Data mining
   - An essential process where intelligent methods are applied to extract patterns or construct models.
3. Pattern/model evaluation
   - To identify the truly interesting patterns or models representing knowledge based on interestingness measures.
4. Knowledge presentation
   - Visualization and knowledge representation techniques are used to present mined knowledge to users.


Knowledge
Pattern/Knowledge Evaluation
Data Mining
Data Preparation
Selection
Data Warehouse
Data Cleaning
Data Integration
Data Sources

Thinking

- What kinds of knowledge to be mined?
- What kinds of databases to work on?
- What kinds of techniques to be utilized?

What kinds of knowledge to be mined?

- characterization: generalize, summarize, contrast
- frequent itemset
- association rules
- classification
- clustering
- trend and deviation analysis
- pattern analysis

...

What kinds of databases to work on?

- Structured data
  - Relational data
  - Transactional data
  - Graph (Network) data
- Unstructured data
  - Textual data
  - Multimedia data (e.g., audio, image, video)
- Mixture of structured and unstructured data

What kinds of databases to work on (cont.)

- Data associated with different applications
  - Sequential data
    - Biological sequences (DNA)
  - Music data
    - Score, MIDI, MusicXML
  - Spatial data
  - Temporal data
  - Social network data

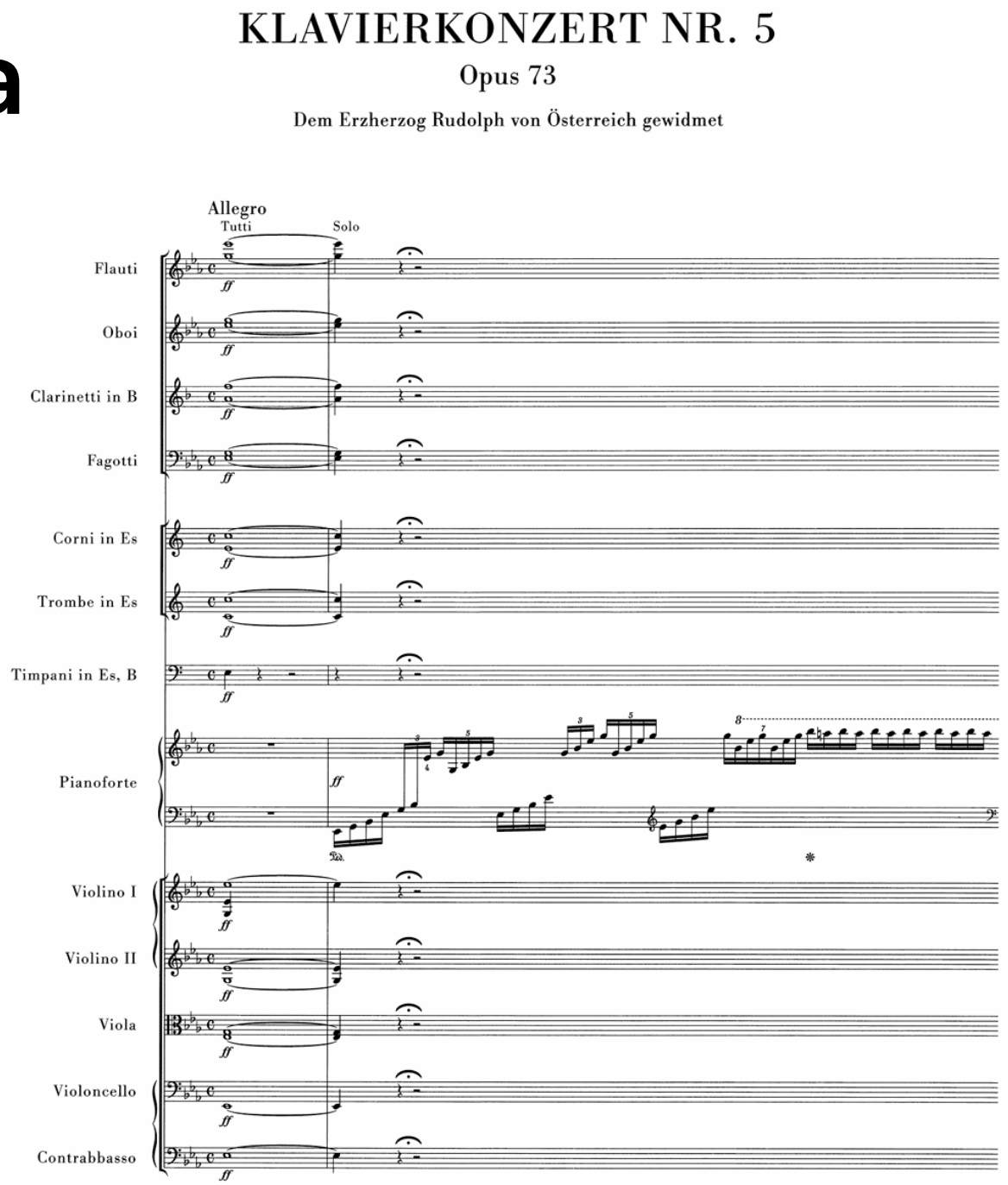
Relational data

customer_id	account_num	lname	fname	mi	address1	addres	adc	address4	city	state_province	postal_code	country
1	87462024688	Nowmer	Sheri	A.	2433 Bailey Road				Tlaxiaco	Oaxaca	15057	Mexico
2	87470586299	Whelply	Derrick	I.	2219 Dewing Avenue				Sooke	BC	17172	Canada
3	87475757600	Derry	Jeanne		7640 First Ave.				Issaquah	WA	73980	USA
4	87500482201	Spence	Michael	J.	337 Tosca Way				Burnaby	BC	74674	Canada
5	87514054179	Gutierrez	Maya		8668 Via Neruda				Novato	CA	57355	USA
6	87517782449	Damstra	Robert	F.	1619 Stillman Court				Lynnwood	WA	90792	USA
7	87521172800	Kanagaki	Rebecca		2860 D Mt. Hood Circle				Tlaxiaco	Oaxaca	13343	Mexico
8	87539744377	Brunner	Kim	H.	6064 Brodia Court				San Andres	DF	12942	Mexico
9	87544797658	Blumberg	Brenda	C.	7560 Trees Drive				Richmond	BC	17256	Canada
10	87568712234	Stanz	Darren	M.	1019 Kenwal Rd.				Lake Oswego	OR	82017	USA
11	87572821378	Murraiin	Jonathan	V.	5423 Camby Rd.				La Mesa	CA	35890	USA
12	87579237222	Creek	Jewel	C.	1792 Belmont Rd.				Chula Vista	CA	40520	USA
13	87587122917	Medina	Peggy	A.	3796 Keller Ridge				Mexico City	Mexico	59554	Mexico
14	87592626810	Rutledge	Bryan	K.	3074 Ardith Drive				Lincoln Acres	CA	30346	USA
15	87597749829	Cavestany	Walter	G.	7987 Seawind Dr.				Oak Bay	BC	15542	Canada
16	87603285908	Planck	Peggy	M.	4864 San Carlos				Camacho	Zacatecas	77787	Mexico
17	87625473141	Marshall	Brenda	S.	2687 Ridge Road				Arcadia	CA	28530	USA
18	87637655735	Wolter	Daniel	P.	2473 Orchard Way				Altadena	CA	49680	USA
19	87650814652	Collins	Dianne		551 Rainier Dr				Oakland	CA	21486	USA
20	87653979700	Baker	Beverly		591 Merriewood Drive				Spring Valley	CA	88762	USA
21	87663244009	Castillo	Pedro		1579 Plaza Rosa				Renton	WA	71442	USA
22	87675641200	Borges	Laurie		1873 Lyon Circle				Bellingham	WA	78588	USA
23	87678398489	Wyro	Shauna	K.	3114 Notre Dame Ave.				La Jolla	CA	27430	USA
24	87681713700	Wyllie	Jacqueline		6318 Marclair Dr.				Santa Fe	DF	99737	Mexico
25	87686740159	Conley	Lin	N.	7814 Milburn Dr.				Haney	BC	65710	Canada
26	87691326700	Bernard	Jose		3928 San Francisco				Burbank	CA	51809	USA
27	87700161933	Macaluso	Charles	B.	5678 McElroy Court				Anacortes	WA	58052	USA
28	87709978818	Wood	Lois	M.	2046 Las Palmas				Everett	WA	88974	USA
29	87717585563	Armstrong	Kevin	A.	608 Lincoln Dr.				Oak Bay	BC	90098	Canada
30	87718932032	Goldey	Cody	A.	7230 Berrellesa Street				Milwaukie	OR	78800	USA
31	87729650139	Nickell	Paula	B.	2144 San Rafael				Concord	CA	63996	USA
32	87738383412	Long	Eric	J.	1676 Greer Ave.				Ladner	BC	99949	Canada
33	87742803692	Solimena	Mary	K.	6843 San Simeon Dr.				Santa Cruz	CA	51307	USA
34	87767067000	Muenich	Nathan		6629 Polson Circle				La Mesa	CA	53422	USA

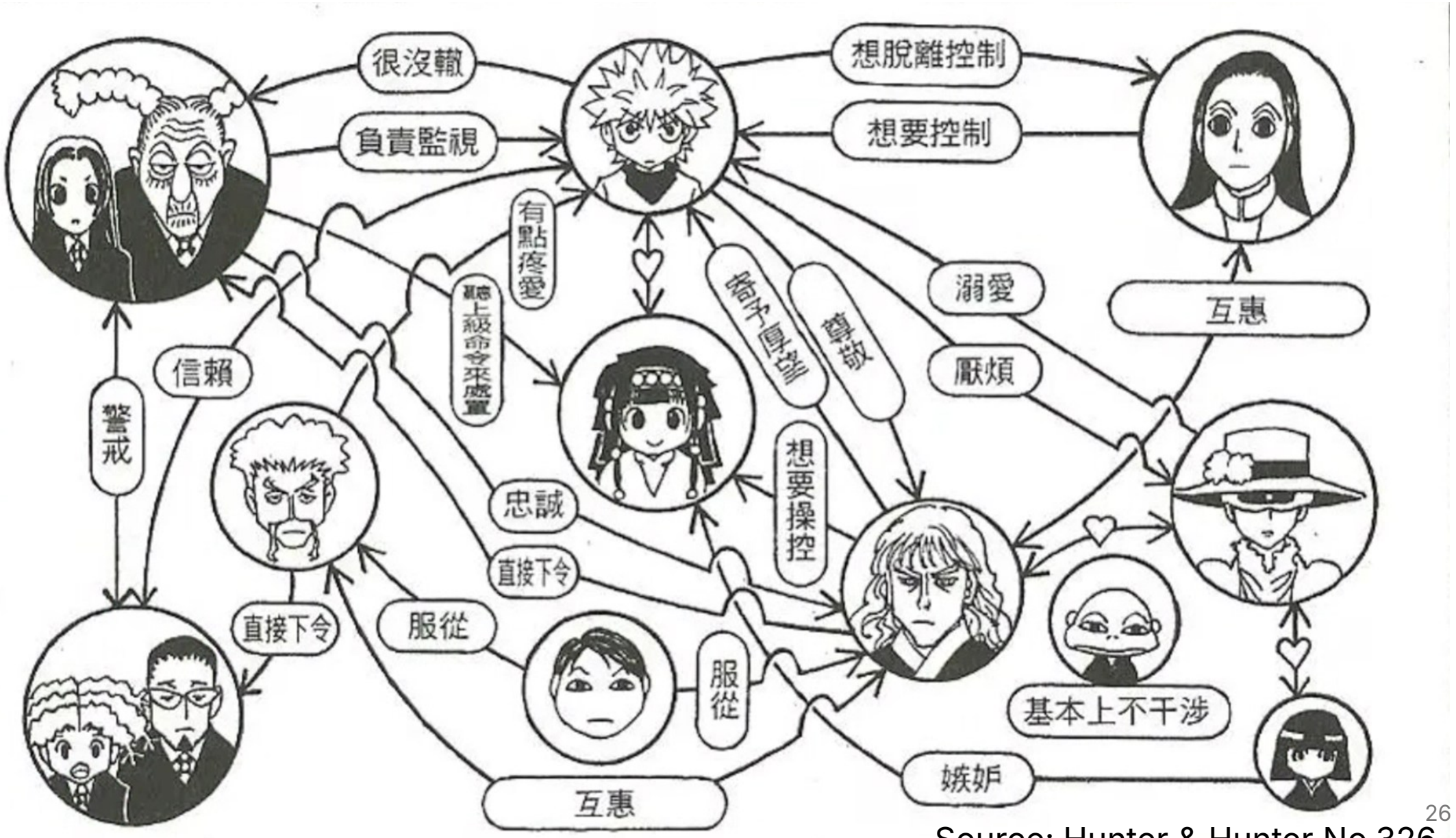
Transaction data

Transaction ID	ITEMS			
1	Bread	Apple	Milk	
2	Beer	Apple	Diaper	
3	Bread	Beer	Apple	Diaper
4	Beer	Diaper		

Music data


KLAVIERKONZERT NR. 5
Opus 73
Dem Erzherzog Rudolph von Österreich gewidmet
Allegro
Tutti
Solo
Flauti
Oboi
Clarinetti in B
Fagotti
Corni in Es
Trombe in Es
Timpani in Es, B
Pianoforte
Violino I
Violino II
Viola
Violoncello
Contrabbasso

Social network data



Source: Hunter & Hunter No.326

Mining various kinds of knowledge

What kinds of techniques to be utilized

descriptive data mining and predictive data mining

Mining Knowledge

- Descriptive data mining
  - Characterize properties of the interested set of data.
  - Find human-interpretable patterns that describe the data.
- Predictive data mining
  - Perform induction on the data set in order to make predictions.
  - Use some variables to predict unknown or future values of other variables.

Mining various kinds of knowledge

Association Rules

Association Rules - Example

- Mining frequent patterns, associations, and correlations
  - Frequent patterns: patterns that occur frequently in data
  - Mining frequent patterns leads to the discovery of interesting associations and correlations within data
    - ➢ What items are frequently purchased together in PX Mart?

Transaction ID	ITEMS			
1	Bread	Apple	Milk	
2	Bear	Apple	Diaper	
3	Bread	Bear	Apple	Diaper
4	Bear	Diaper		

Association Rules – Example (cont.)

- Frequent Itemset (k-itemset)
  - {Beard, Apple}: 50% (Transaction 1, 3)

Transaction ID	ITEMS			
1	Bread	Apple	Milk	
2	Beer	Apple	Diaper	
3	Bread	Beer	Apple	Diaper
4	Beer	Diaper		

Association Rules – Example (con.t)

- Frequent Itemset
  - {Beard, Apple}: 50% (Transaction 1, 3)
  - {Beer, Apple, Diaper}: 50% (Transaction 2, 3)

Transaction ID	ITEMS			
1	Bread	Apple	Milk	
2	Beer	Apple	Diaper	
3	Bread	Beer	Apple	Diaper
4	Beer	Diaper		

Association Rules – Example (con.t)

- Frequent Itemset
  - {Beard, Apple}: 50% (Transaction 1, 3)
  - {Beer, Apple, Diaper}: 50% (Transaction 2, 3)
  - {Beer, Diaper}: 75% (Transaction 2, 3, 4)

Transaction ID	ITEMS			
1	Bread	Apple	Milk	
2	Beer	Apple	Diaper	
3	Bread	Beer	Apple	Diaper
4	Beer	Diaper		


尿布
啤酒

Association Rules – Example (con.t)

- Strong Rules
  - Beer ^ Diaper → Apple (2/3)
  - Beer ^ Apple → Diaper (2/2)
  - Apple → Bread (?)

Transaction ID	ITEMS			
1	Bread	Apple	Milk	
2	Beer	Apple	Diaper	
3	Bread	Beer	Apple	Diaper
4	Beer	Diaper		

Association Rules – Example (con.t)

- Strong Rules
  - Beer ^ Diaper → Apple (2/3)
  - Beer ^ Apple → Diaper (2/2)
  - Apple → Bread (2/3)
  - Bread → Apple (?)

Transaction ID	ITEMS			
1	Bread	Apple	Milk	
2	Beer	Apple	Diaper	
3	Bread	Beer	Apple	Diaper
4	Beer	Diaper		

Association Rules – Example (con.t)

- Strong Rules
  - Beer ^ Diaper → Apple (2/3)
  - Beer ^ Apple → Diaper (2/2)
  - Apple → Bread (2/3)
  - Bread → Apple (2/2)

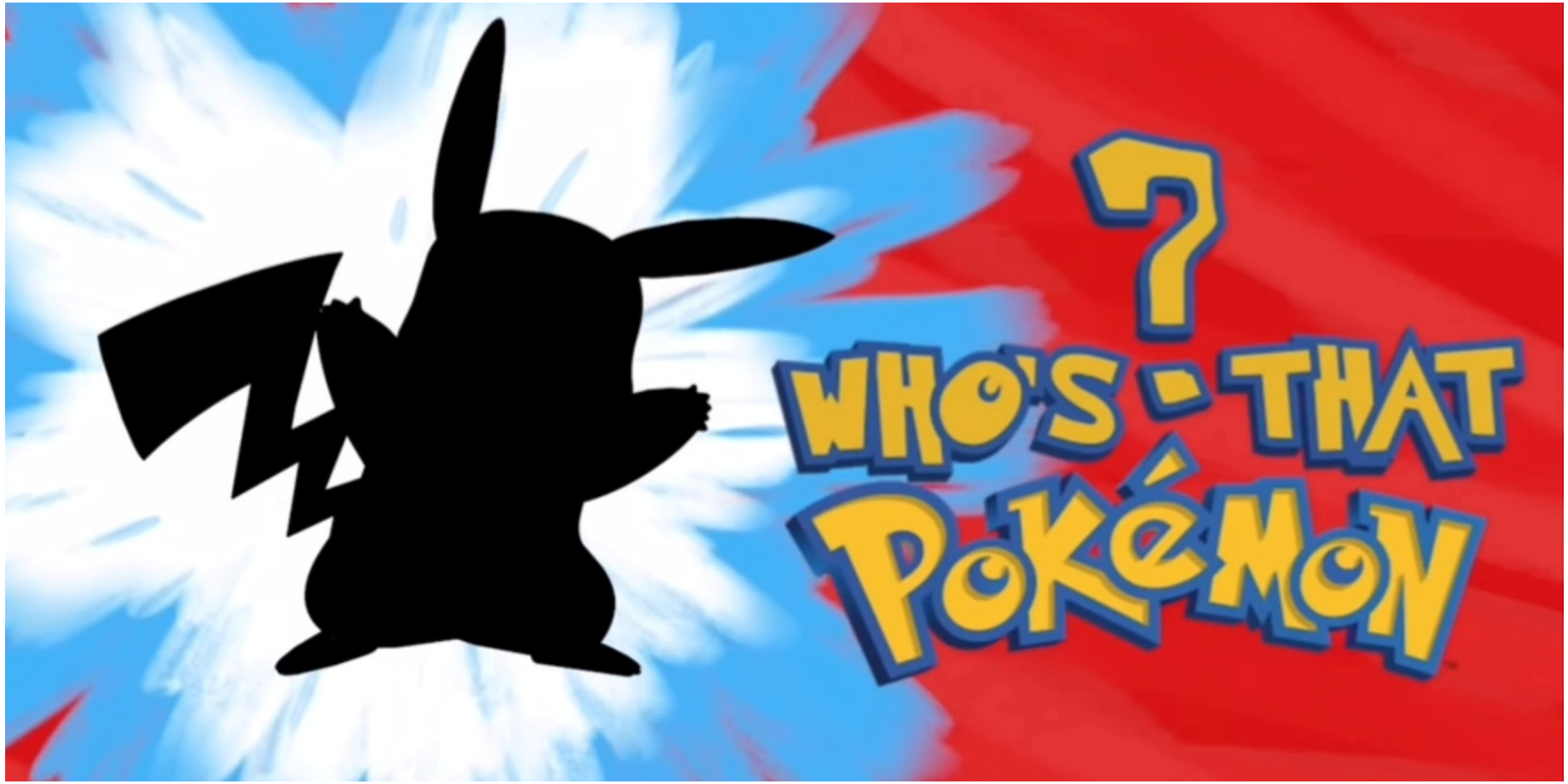
Confidence
Support

Transaction ID	ITEMS			
1	Bread	Apple	Milk	
2	Beer	Apple	Diaper	
3	Bread	Bear	Apple	Diaper
4	Beer	Diaper		

Mining various kinds of knowledge

Classification

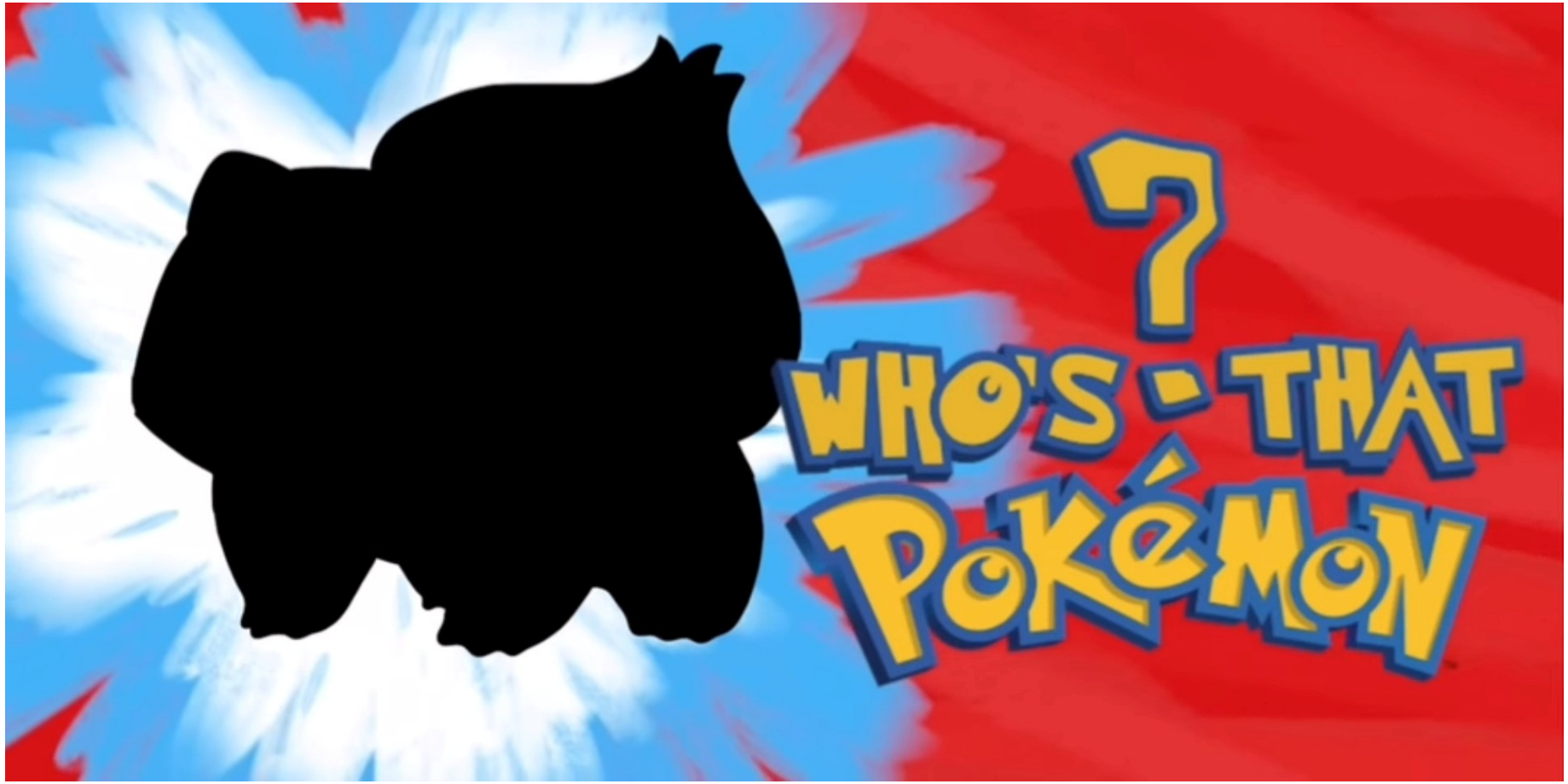
Classification



Classification (cont.)



Classification (cont.)

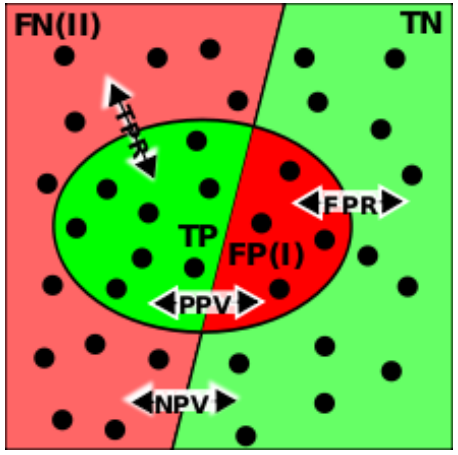


Classification (cont.)



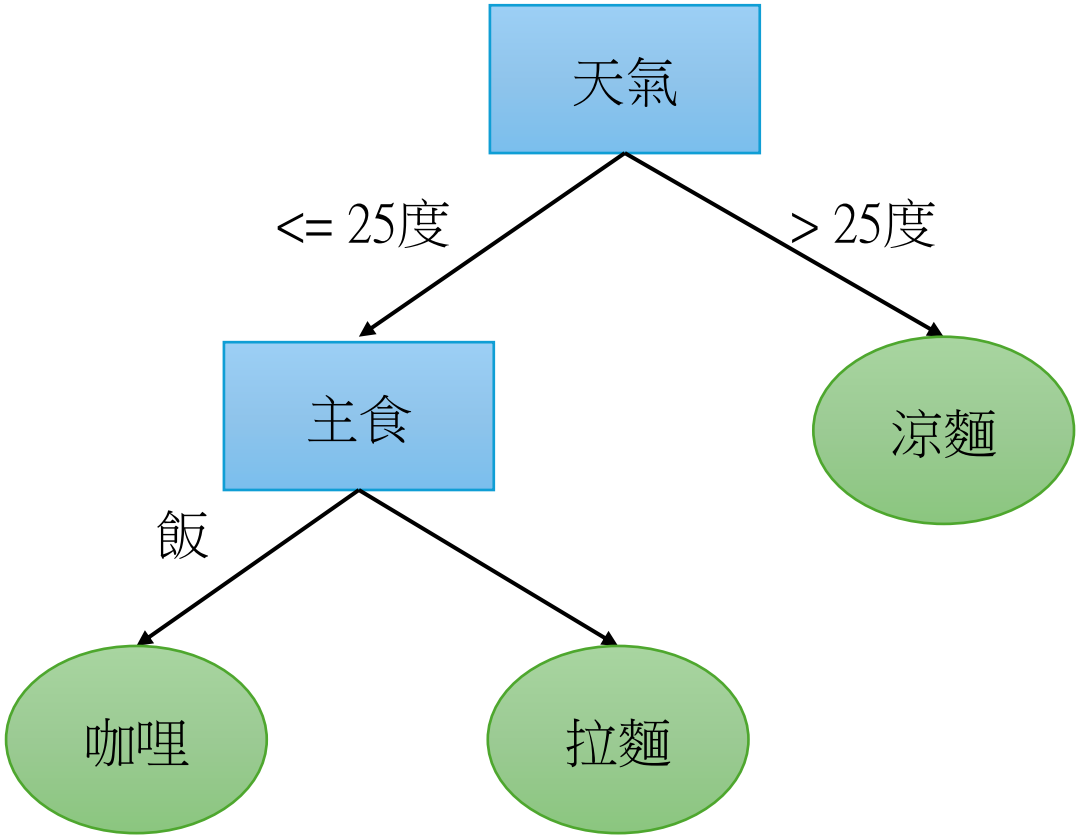
Classification (cont.)

- Classification is the process of finding a model (or function) that describes and distinguishes data classes or concepts.
  - Supervised Learning
- The model is derived based on the analysis of a set of training data (i.e., data objects for which the class labels are known).
- Organizes data into given classes based on attribute values.



Classification Example (Decision Tree)



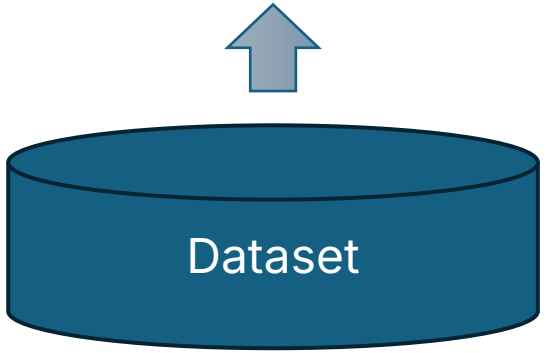


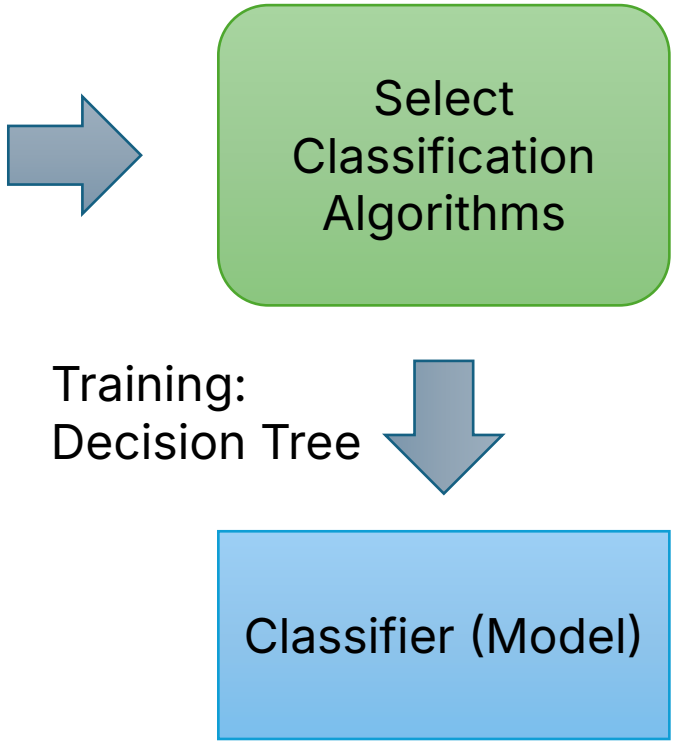
Classification Process

- Step 1. Training

	Position	YEARS	Tenured
Bob	Assistant Prof.	2	NO
Alice	Associate Prof.	8	YES
John	Prof.	3	YES
Amy	Associate Prof.	9	YES
Mike	Assistant Prof.	5	NO

Training data





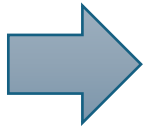
Classification Process

- Step 1. Training

	Position	YEARS	Tenured
Bob	Assistant Prof.	2	NO
Alice	Associate Prof.	8	YES
John	Prof.	3	YES
Amy	Associate Prof.	9	YES
Mike	Assistant Prof.	5	NO

Training data

Tenured is YES
If Position == “Professor”
OR
YEARS >= 8



Select Classification Algorithms

Training: Decision Tree



Classifier (Model)

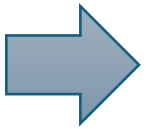
Classification Process (cont.)

- Step 2: Evaluation & Inference

	Position	YEARS	Tenured
Jessica	Associate Prof.	2	NO
Kevin	Assistant Prof.	15	YES
Frank	Associate Prof.	3	NO
Linda	Prof.	10	YES
Mark	Prof.	5	YES

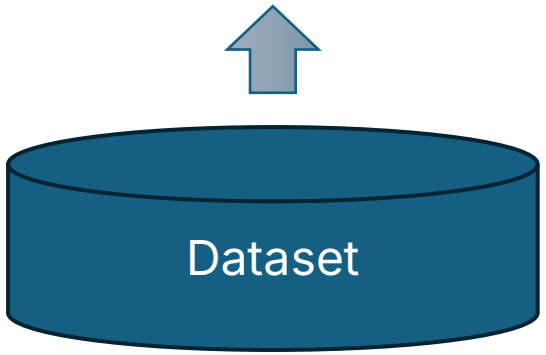
Test data

Tenured is YES
If Position == “Professor”
OR
YEARS >= 8

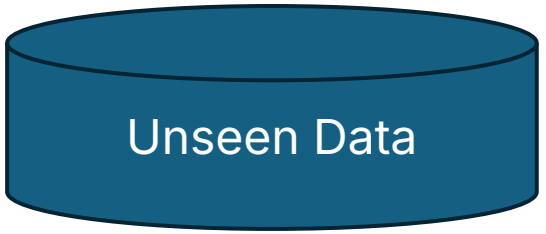


Classifier (Model)





Training data
vs
Test data



Mining various kinds of knowledge

Clustering

Clustering

- Unsupervised classification
- Organizes data into classes based on attribute values

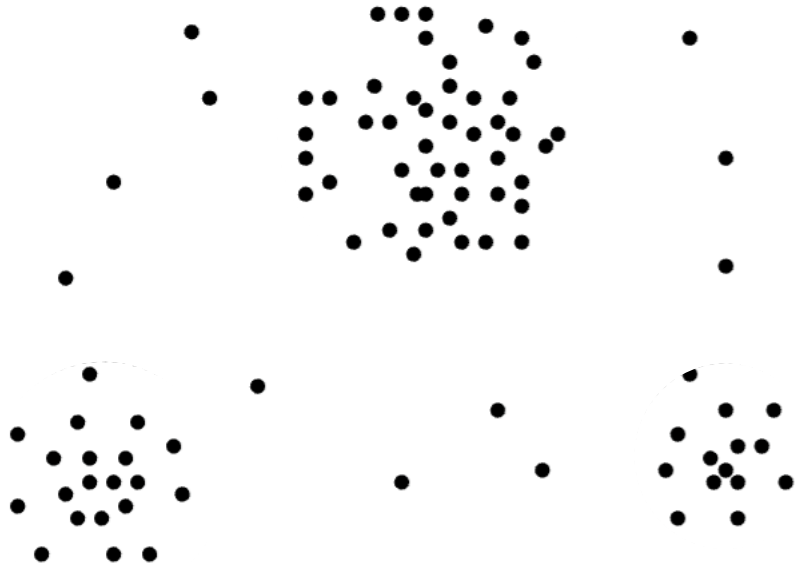


FIGURE 1.3

A 2-D plot of customer data with respect to customer locations in a city, showing three data clusters.

Clustering (cont.)

- Unsupervised Learning
- Organizes data into classes based on attribute values.

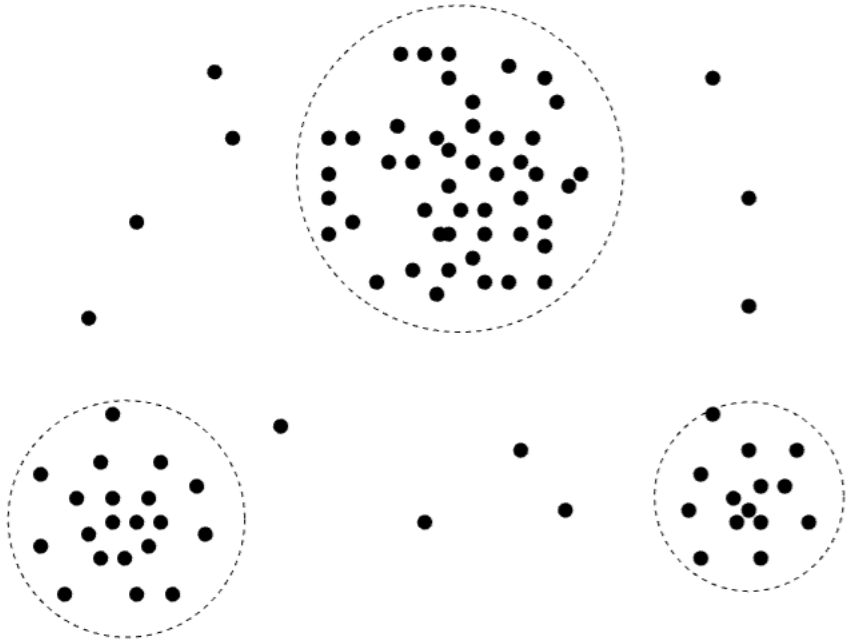


FIGURE 1.3

A 2-D plot of customer data with respect to customer locations in a city, showing three data clusters.

Clustering (cont.)

- Group data objects without consulting class labels.
- Clustering can be used to generate class labels for a group of data.
- Objects are clustered or grouped based on
  - Maximizing the intraclass similarity.
  - Minimizing the interclass similarity.

Semi-supervised learning

- Labeling data is often expensive.
- Unlabeled data is easy to collect.
- Pseudo-labeling (self-training)
  - The model predicts labels for unlabeled data, then trains on those predictions as if they were true labels.
  - Clustering methods

Mining various kinds of knowledge

Outlier Detection

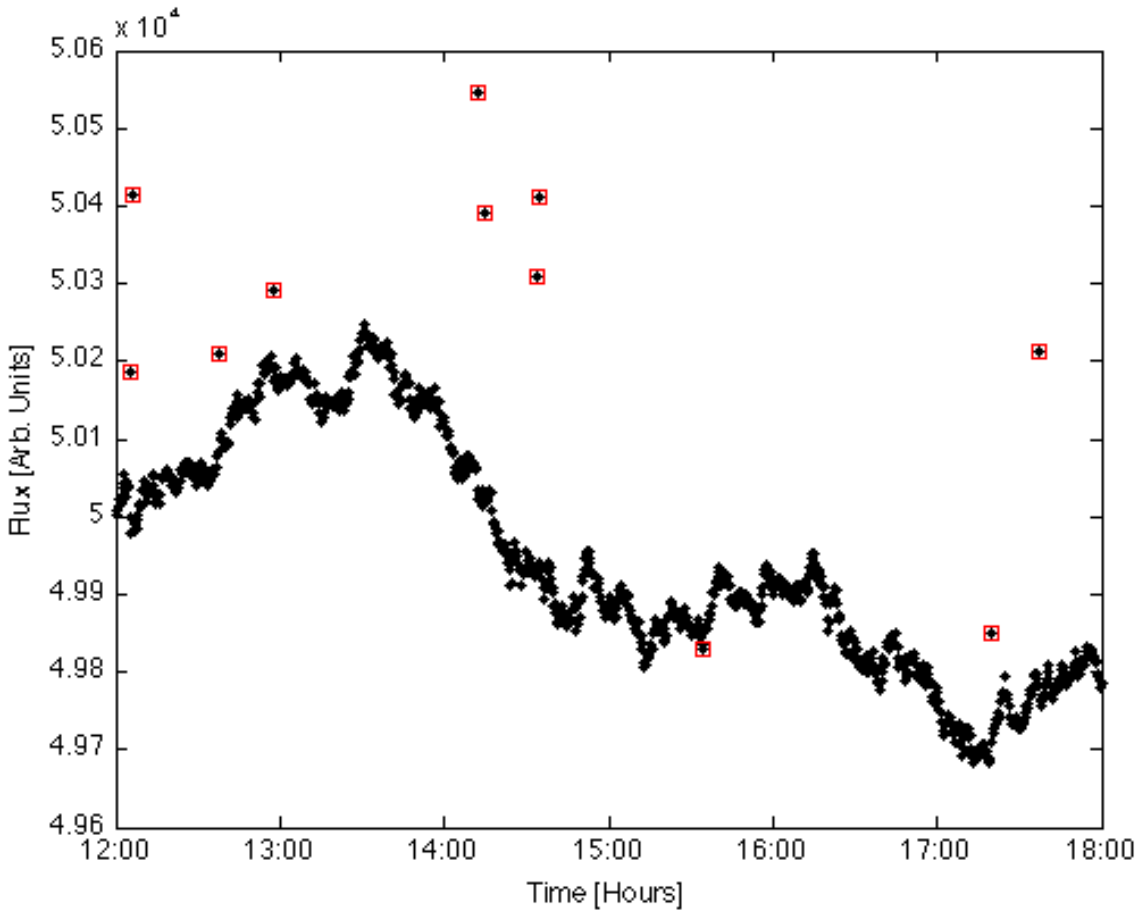
Outlier Detection

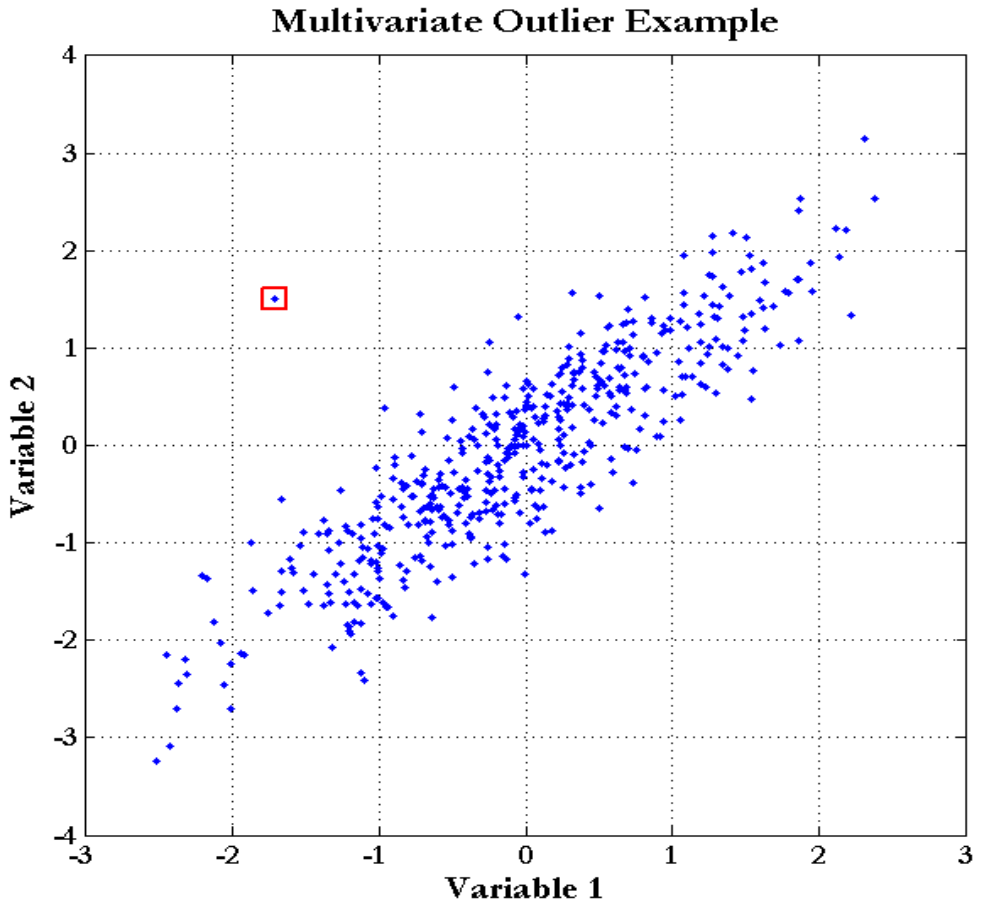
- A data set may contain objects that do not comply with the general behavior or model of the data.
  - These data objects are outliers
- Many data mining methods discard outliers as noise or exceptions.
  - Outlier analysis
  - Anomaly mining
  - Anomaly detection
- Noise or exception?
  - One person’s garbage could be another person’s treasure



Outlier Detection

- Useful in fraud detection, rare events analysis





Mining various kinds of knowledge

Deep Learning

Deep Learning

- Deep learning: A fast expanding dynamic frontier in machine learning .
- Deep learning has developed various neural network architectures:
  - Feed-forward neural networks
  - Convolutional neural networks
  - Recurrent neural networks
  - Graph neural networks
  - Transformer

Deep Learning

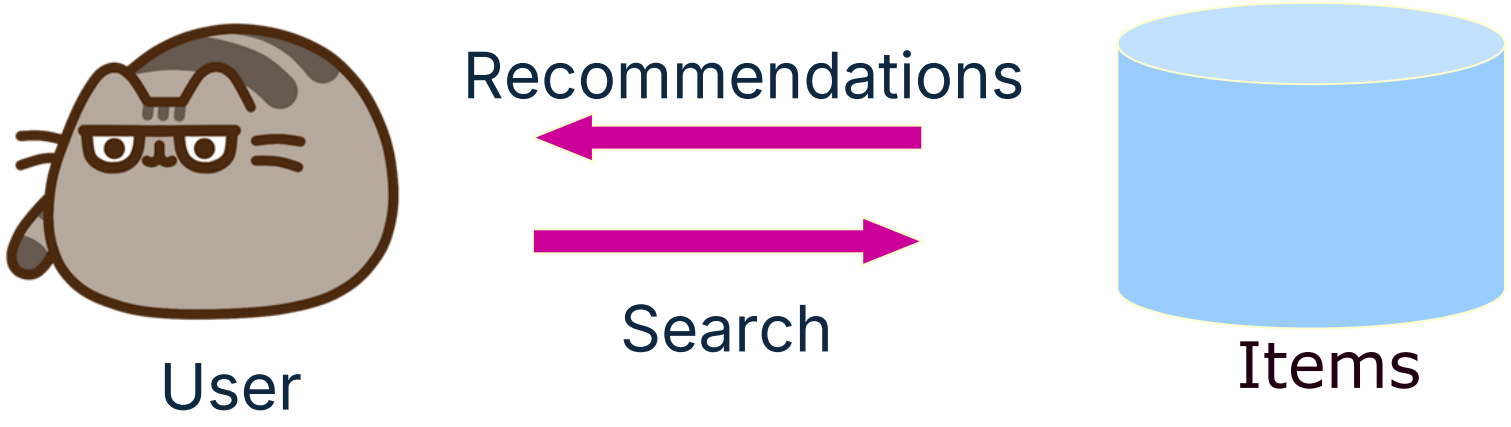
- Deep learning has broad applications in computer vision, natural language processing, machine translation, social network analysis, and so on.
- Deep learning has been reshaping a variety of data mining tasks.
  - Ex. classification, clustering, outlier detection, etc.

Mining various kinds of knowledge

Recommendation

Recommendation

- Recommendation is a suggestion given to help someone choose the best option based on their needs, preferences, or past behavior.
  - Content-based
  - Collaborative filtering
- Cold start user recommendation



Music, products, web sites, blogs, news items, ...

Mining various kinds of knowledge

Social Network Mining

Social Network Mining

- Analyze social networks (graphs) to discover patterns, relationships, and useful knowledge about people (or entities) and their connections.
  - Community detection
  - Centrality analysis
  - Link prediction
  - Node classification
  - Anomaly detection

Mining various kinds of knowledge

Text Mining

Text Mining

- Text mining is the process of using computer to extract useful information, patterns, or knowledge from text data.
  - Keyword extraction: TF-IDF
  - Topic modeling: Latent Dirichlet Allocation
  - Information extraction
  - Summarization
- Language Model (LM) is one of the major approaches to advancing language intelligence of machines.

Summary

Knowledge Discovery (KDD) Process

1. Goal: Learning the application domain
2. Data:
   a) Gathering & integrating data
   b) Cleaning & preprocessing data
   c) Reducing & projecting data (useful features, dimensionality /variable reduction)
3. Method
   a) Choosing functions of data mining
   b) Choosing the mining algorithm
   c) Data mining
   d) Evaluation results
4. Interpretation (visualization, alteration, removing redundant patterns)
5. Use of discovered knowledge

Evaluation of Knowledge

- Are all mined knowledge interesting?
  - One can mine tremendous amount of “patterns”
  - Some may fit only certain dimension space (time, location, ...)
  - Some may not be representative, may be transient, ...
- Evaluation of mined knowledge → directly mining only interesting knowledge?

What’s Interesting?

- Data mining has the potential to generate a lot of results
  - Are all of the mining results interesting?
  - Can a data mining system generate all of the interesting patterns?
  - Can a data mining system generate only interesting patterns?

① Easily understood by humans

② Valid on new or test data with some degree of certainty

③ Potentially useful

④ Novel

Data Mining Issues

- Overfitting
- Outliers (Noisy Data)
- Interpretation
- Visualization
- High Dimensionality
- Irrelevant Data / Missing Data
- Privacy

Data Mining and Applications

- Web page analysis: classification, clustering, ranking
- Collaborative analysis & recommender systems
- Basket data analysis to targeted marketing
- Biological and medical data analysis
- Data mining and software engineering
- Data mining and text analysis
- Data mining and social and information network analysis